06-WIRELESS-AND-IOT-MODULES

Item #	Module Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**NRF24L01+PA+LNA Wireless RF Transceiver Module Long Range 1100M with SMA Antenna (2-Pack)**	Long-range 2.4 GHz digital packet radio transceiver with Power Amplifier (PA) + Low Noise Amplifier (LNA) and **external SMA duck antenna**; 2 Mbps low power signal amplifier; up to 1,100 metres line-of-sight range; SPI bus; for Uno, Arduino & ESP32 wireless communication projects	**5 Packs (10 Units)**	**Consumable**
02	**Bluetooth Low Energy (BLE) Module (HM-10 / AT-09)**	2.4 GHz Bluetooth 4.0/5.0 BLE transceiver; UART serial; compatible with **both Android and Apple iOS smartphones**; 3.3V/5V tolerant	**10 Units**	**Consumable**
03	**Bluetooth Serial Module (HC-05)**	2.4 GHz Bluetooth V2.0+EDR classic serial module; SPP serial profile; UART AT command set; configurable Master/Slave mode; 3.3V logic (5V VCC tolerant); legacy Android/PC serial telemetry	**2 Units**	**Consumable**
04	**433 MHz / 315 MHz RF Transmitter & Receiver Pair**	ASK wireless super-regenerative RF transmitter and receiver pair; 315/433 MHz ISM band; operating voltage 3.3V–5V DC; range 20–100m for basic wireless telemetry and remote control experiments	**2 Pairs**	**Consumable**
05	**GSM / GPRS Cellular Module (SIM900)**	Quad-Band 850/900/1800/1900 MHz cellular shield; onboard standard SIM socket; UART serial AT command control; external antenna; 2G cellular SMS and telemetry demonstration	**2 Units**	**Consumable**
06	**RFID Reader Module & Tag Kit (RC522)**	13.56 MHz contactless RFID/NFC reader board (MFRC522); SPI bus; includes 1x RFID contactless card and 1x key fob tag	**10 Kits**	**Consumable**
07	**GPS Satellite Module (u-blox GY-NEO6MV2 / NEO-8M)**	High-sensitivity satellite receiver module with active ceramic patch antenna; EEPROM config retention; 9600 baud serial NMEA output	**3 Units**	**Consumable**